Weekly Report

Name: Digvijay Basiya
Project: UAV-Based Traffic Safety & Video Analytics
Week: 21 Mar 2026 – 28 Mar 2026
Submission Date: 28 Mar 2026

Detailed Summary of Work Done During Current Week

A. Zone Consistency and Validation

- Performed additional validation of predefined zones across extended video segments.
- Ensured that zone boundaries remain consistent and aligned with the road structure throughout the video.
- Verified that zones can reliably serve as reference regions for analyzing vehicle movement patterns.

B. Semi-Automated Annotation Pipeline Enhancement

This week focused on stabilizing and improving the annotation pipeline for large-scale UAV video data.

1. Inference Output Refinement

- Executed object detection inference on longer video segments.
- Verified consistency of detection outputs across frames.
- Analyzed detection quality to ensure suitability for annotation pre-processing.

2. CSV Data Validation

- Validated structured CSV outputs for correctness and completeness.
- Ensured uniform formatting of:
  - Frame number
  - Bounding box coordinates
  - Class labels
  - Confidence scores
- Improved data consistency to support reliable XML conversion.

3. CSV to XML Conversion Stabilization

- Further refined CSV-to-XML conversion script.
- Verified compatibility of generated XML files with CVAT annotation format.
- Successfully tested annotation loading for larger frame sequences in CVAT.

4. CVAT Annotation Workflow Improvement

- Continued testing annotation correction workflow in CVAT.
- Improved efficiency of refining bounding boxes instead of manual annotation.
- Evaluated usability of the pipeline for handling extended video durations.

C. Vehicle Trajectory Extraction Advancement

- Extended trajectory extraction to longer frame segments.
- Generated more continuous vehicle paths using centroid-based calculations.
- Visualized trajectories to verify movement continuity and correctness.
- Identified improvement areas in maintaining consistent vehicle identity across frames.

D. Pipeline Integration Progress

- Worked towards integrating detection, annotation, and trajectory extraction into a unified workflow.
- Ensured smooth data flow between CSV, XML, and visualization stages.
- Organized intermediate outputs for better tracking and reproducibility of results.

Tasks Planned for Coming Week

A. Annotation Workflow Finalization

- Complete stabilization of CSV → XML → CVAT pipeline.
- Perform annotation validation on larger portions of the dataset.
- Improve scalability of the annotation process.

B. Trajectory Optimization

- Improve consistency of vehicle identity across frames.
- Enhance trajectory smoothness and continuity.
- Validate trajectory accuracy across longer sequences.

C. Behavior Analysis Preparation

- Begin mapping vehicle trajectories relative to predefined zones.
- Prepare structured outputs for detecting movement patterns.
- Continue documentation for final project reporting.